

Findings on AGSD architectural knowledge sharing

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Abstract—Agile Software Development (ASD) is increasingly common in globally distributed teams – it is known as AGSD. However, while in ASD face-to-face interactions are preferred over document-based communications, in global software development document-based communication is preferred. These differences affect Architectural Knowledge Management. AGSD workers preferably convey architectural knowledge (AK) by frequent interactions across sites through unstructured and textual electronic media (UTEM) (chats, emails, forums, etc.), i.e., AK is articulated in these media. UTEM leave unstructured logs of shared AK. We present an empirical study to understand AK articulation through UTEM in AGSD teams. Our results are an ontological model about AK articulation through UTEM in AGSD teams, and 11 categories of UTEM interactions across sites, where requirements and coding themes were prominent. Also, we found that AK in UTEM is perceived as important, regardless of the interaction frequency.

Keywords— agile global software development; virtual teams; architectural knowledge; unstructured and textual electronic media.

I. INTRODUCTION

Nowadays, the adoption of agile software development (ASD) in global software development (GSD) companies is a fact [1], i.e., they use an agile global software development (AGSD) paradigm. In AGSD, communication between local and remote members is a challenge since ASD and GSD have different approaches. In ASD people interactions are preferred over following processes, and working software is preferred over comprehensive documentation [2]; in GSD geographic distribution of people increases the need for more explicit communication [3]. Knowledge management (KM) is a significant challenge in AGSD [4], due to a limited knowledge transfer [5]. Hence, architectural knowledge management (AKM) is also a challenge in AGSD, since ASD reduces the time to write proper documentation, which negatively affects the coding phases [6], and generates documentation debt [7].

In ASD, face-to-face (f2f) communication is preferred, thus, documentation debt is solved with teammates' interaction. AGSD developers prefer text media (particularly instant messaging - IM) because they reduce communication problems caused by language differences [8]. Text media such as email, IM, forums, etc., usually leaves a textual record of users'

interactions. In these media (unstructured textual electronic media – UTEM), there will be architectural knowledge (AK) recorded that will not likely be recorded elsewhere, i.e., tacit AK is articulated in UTEM. These records could be valuable to complement and contextualize informal and ad-hoc documents used in AGSD. However, to the best of our knowledge, AK articulation in UTEM has not been sufficiently studied, and it is unclear which AK aspects are shared.

This work presents the results of an empirical study conducted in four Mexican AGSD companies. We observed UTEM interactions between local and remote members, and local members f2f interactions, where AK issues were addressed. Our main goal was to understand and characterize AK articulation through UTEM in an AGSD environment.

II. STUDY DESCRIPTION AND RESULTS

We conducted a two-phase study: (1) observation and (2) exploration, with 20 software engineers of four AGSD Mexican companies (named A, B, C and D – 11, 4, 3, and 2 participants respectively). These companies had projects in the US, Spain and Latin America, and all the participants have contact with remote counterparts through UTEM. The companies apply Scrum to manage projects, and different software architecture practices (e.g. spike solutions, agile and ad-hoc modeling, participative design, pair programming).

In the observation phase, we stayed in the facilities of company A (the only one who granted us permission to monitor and analyze their UTEM interactions) during 3 weeks, 5 days a week, and 8 hours per day, to observe UTEM interactions among remote and local team members, f2f interactions among local personnel, and documents related to SA topics. We observed f2f and SA documentation, to find a relation with UTEM interactions. We registered f2f interactions in a log with the following fields: beginning and ending time, support media (computer, notebook, board, etc.), participant name and a brief description of the interaction topic. UTEM interactions were captured with a custom developed software tool, which connects to Microsoft Lync, Microsoft Exchange and discussion forums of Target Process tool. We used the 4+1 views architectural model [9] as a reference to identify when a document or an interaction relates to a SA topic. The findings of this phase can be summarized as a UTEM interactions categorization, a list of artifacts used to record architectural issues, and the main intentions of f2f interactions. In the

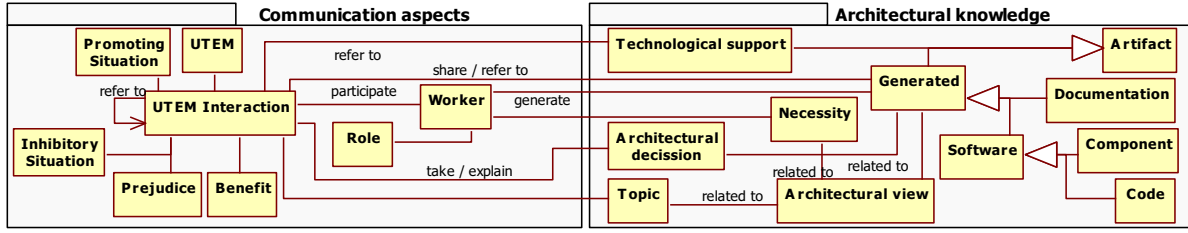


Fig. 1. Ontology that represents the involved aspects in AK articulation through UTEM in AGSD teams (all relations without label are of type “have”).

exploration phase, we exposed the findings of the on-site observation to all the participants in four focus group sessions (one per company), aiming at finding correspondences among the companies regardless of their work dynamics. The audio of the focus groups sessions was recorded and transcribed. Then, we applied Grounded Theory to understand AK articulation through UTEM in AGSD teams.

As a result of the observation phase, we found that most of the architectural documents used by the participants were related to requirements specification (scenarios view). Technical documentation is almost absent and it is usually done by specific request only. Regarding the SA interactions, we registered 45 f2f interactions and 216 UTEM interactions (145 IM conversations, 51 email messages and 20 discussion forums comments). It is worth saying that MS Lync defines an IM conversation as the group of messages sent while an IM window is opened. Based on these registered interactions in company A, and on the focus group sessions with all the participating companies, we developed an ontology that represents our understanding of AK articulation through UTEM in AGSD teams (see Fig. 1). The resulting ontology has two parts: communications aspects and AK aspects.

As the next step, we analyzed the captured UTEM interactions in order to categorize the AK topics that are discussed in these media. We obtained 11 UTEM interaction categories and they were related to different architectural views of the 4+1 views model. We observed that issues related to scenarios and implementation views were prominent. All participants evaluated the obtained categories in terms of incidence and importance, during the focus group sessions. We found that the perceived importance of the evaluated topics is significantly higher than its perceived incidence. Regarding f2f interactions, we observed that 55% were related to the scenarios view, and the rest were related to the implementation and deployment views. We may see a correspondence between UTEM and f2f interactions, as scenarios and implementation views were frequent in UTEM interactions. Also, given that observation was performed only in company A, during the focus group sessions we asked the rest of participants for their most frequent architectural topics in f2f interactions. The topics reported were consistent with those observed in company A.

III. DISCUSSION AND CONCLUSIONS

Despite that the only SA documents found were related to the scenarios view, we found that the most frequent SA topic in f2f interactions was also related to the scenarios view. This is due to the scenarios documents being generally incomplete or unclear. Additionally, f2f interactions related to the implementation view were the second in frequency. This

reflects at least two agile values in AGSD teams: “Individuals and interactions over processes and tools” and “Working software over comprehensive documentation”.

Further, most of the identified categories of UTEM interactions were related to the scenarios and implementation views, just as for the registered f2f interactions. This behavior corresponded with the usual behavior in ASD, i.e., the analysis and design phases are diluted, because problems are generally understood by experimenting with code [10]. Also, we found that AK shared through UTEM is important, regardless of the interaction frequency in AGSD teams. These findings provide evidence of the importance of UTEM interactions, and that AK is shared through these type of media.

Based on our ontological model about AK articulation through UTEM in AGSD teams, and the other findings of the study, we observe a trend where ASD practices are prominent in an AGSD. In fact, the most important drivers for AGSD relate to the Agile Principles and the rest of the drivers relate to global business perspectives [5].

As future work, we plan to formalize our understanding of AK articulation through UTEM in terms of knowledge management, aiming at obtaining a set of design guidelines that support access to AK in UTEM through a software tool.

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